

Operational Contributions and Evidence Profiles

Contribution-level operational and evidence profile of data-dependent stopping procedures

This table reports 95 operational contributions from 54 primary studies. Evidence states describe what was documented for each contribution; they do not form a readiness score, ranking, certification, or universal recommendation. Symbols: ✓ demonstrated evidence; ~ limited or partial evidence; — evidence not reported. Robustness is assessed for non-normality, autocorrelation/dependence, nonstationarity, and model misspecification. Family 1 subfamilies follow the inferential-target partition used in the manuscript; Cross-cutting identifies contributions spanning multiple targets.

Table 1: Operational Contributions and Evidence Profiles. Contribution-level operational and evidence profiles.

ID	Study	Family	Subfamily	Operational contribution and branches	Consider when	Finite-sample evidence	Alternative comparison	Robustness	Computational / scalability evidence	Implementation availability
01	Alexopoulos et al. (2015)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequest for stationary-quantile CIs; no-precision, relative-precision, and absolute-precision branches	Stationary quantile; relative/absolute precision	~ 1,000 replications in three processes and several quantiles/configurations cover the no-precision and relative branches, but not the absolute branch	— None reported.	— Non-normality. ✓ Autocorrelation/dependence: AR(1) and queueing processes contrast dependence and autocorrelation. — Nonstationarity. — Model misspecification.	✓ batch/sample sizes reported in multiple processes, quantiles, and precisions	~ detailed formulas and steps, but no counter, NR semantics, or code
02	Bandyopadhyay et al. (2015)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Fully sequential fixed-width CI rule; RPW-target and BB-type branches	Mean difference; absolute precision	✓ 10,000 replications on binary and normal grids, with coverage and E(N); there is undercoverage and one impossible cell	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ E(N) evaluated across multiple parameters, widths, and acquisition policies	~ rule/formulas reproducible in principle, but no executable code/pseudocode and no fallback for invalid states
03	Bandyopadhyay et al. (2015)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage Stein-type procedure; C1-C3 branches and two stage-1 acquisitions	Mean difference; two-stage absolute precision	✓ C1-C3 evaluated by simulation in RPW-target and BB-type configurations, with coverage and E(N_T)	✓ C1-C3 are compared in a common scenario; the comparison is restricted to tabulated configurations	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ E(N_T) reported for distinct calibrations and configurations	~ horizon and updates are specified, but there is no code and fallbacks for invalid counts/variances are missing
04	Semelhago et al. (2022)	Family 5 - Resource-aware acquisition and stopping		Parallel rGMIA package: q-CEI, closed form, greedy selection, and MATLAB OOP implementation; stopping by precision or budget	Parallel acquisition; precision/budget	✓ 50 macro-replications, q=2-5, different screenings, and gap/cost metrics; terminal decision is not directly validated	✓ greedy batch q-CEI compared with largest marginal CEIs in a common environment	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ times measured systematically for several q and screening sizes	~ GitHub repository and four-class MATLAB architecture are reported, but access/usability were not verified in this reading
05	Silva et al. (2023)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Ideal exact sequential procedure for a fixed-width or fixed-accuracy interval	Population size; fixed width or multiplicative accuracy	— None reported.	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ described as computationally intensive, without systematic scaling evaluation	~ formulas and MLE search are supplied, but Eq. (7) is not literally executable without an interpretive decision
06	Silva et al. (2023)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	MRM Monte Carlo for a fixed-width or fixed-accuracy interval	Population size; Monte Carlo fixed width or accuracy	✓ coverage and withdrawals evaluated on grids of N, t, d, or δ ; results include exceptions to the effort advantage	✓ MRM compared with AS in common scenarios by coverage and effort	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ recapture effort quantified across multiple scales/configurations; Monte Carlo computational cost is discussed only qualitatively	~ Eq. (9) and mechanism are supplied, but m, code, software, seeds, and a rule for decisions near α are missing
07	Alexopoulos et al. (2016a)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequem for extreme quantiles, with no-precision, absolute-precision, and relative-precision branches	Extreme quantile; relative/absolute precision	✓ 1,000 replications in three processes, quantiles, and finite regimes; coverage and size reported	~ Comparisons with Sequest/other published numbers are not uniform across all configurations	✓ Non-normality: Multiple nonnormal marginal distributions were evaluated. ✓ Autocorrelation/dependence: Processes with distinct dependence structures were compared. — Nonstationarity. — Model misspecification.	✓ Sample cost quantified in multiple processes/quantiles, up to about 38 million; CPU/memory not measured	~ Detailed algorithm and formulas; Java mentioned, but no verifiable artifact
08	Alexopoulos et al. (2016b)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	SPSTS for stationary-mean CI, with relative, absolute, and no-precision branches; optional post-stopping σ^2 CIs	Stationary mean; relative/absolute precision	✓ Five processes and multiple precision regimes with coverage, half-width, and sample size	~ ASAP3 and Skart are identified comparators, but some results come from a previous study and the design is uneven	✓ Non-normality: ARTOP and queueing processes include nonnormal marginal distributions. ✓ Autocorrelation/dependence: AR(1), queueing processes, and ARTOP contrast dependence structures. — Nonstationarity. — Model misspecification.	✓ Sample size compared across several processes/regimes; overall time remains open	~ Formulas, algorithm, and flowchart allow partial reconstruction, but executable states are unspecified and no package was located
09	Alexopoulos et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Automated Sequem for extreme quantiles; upper no-precision/precision branches and an intended lower counterpart	Extreme quantile; upper precision branch	✓ Seven processes, three quantiles, several precision levels, and 1,000 replications per configuration	✓ Direct comparison with Sequest under common configurations and metrics	✓ Non-normality: Seven processes include distinct nonnormal marginal distributions. ✓ Autocorrelation/dependence: Seven processes include distinct dependence structures. — Nonstationarity. — Model misspecification.	✓ Sample, observed time, and sorting cost are reported across multiple configurations, although without global algorithm analysis	~ Java implementation reported and procedure detailed, but code only promised and no verifiable artifact
10	Alexopoulos et al. (2017)	Family 4 - Operational-state sufficiency		Ad hoc Sequem extension for non-extreme quantile CIs using already acquired data	Non-extreme quantile; post-run CI	~ One detailed study and claims without complete results for additional applications; nonmonotone precision and overcoverage	— None reported.	~ Non-normality: One case is reported in detail; additional results are omitted. ~ Autocorrelation/dependence: One case is reported in detail; additional results are omitted. — Nonstationarity. — Model misspecification.	— None reported.	~ Procedure described by reference to Sequest steps, without verifiable artifact
11	Alexopoulos et al. (2018)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequest for non-extreme quantiles, with absolute/relative precision and real-time/file modes	Non-extreme quantile; relative/absolute precision	✓ 1,000 replications in two queues, several quantiles, and two precision regimes, with coverage and post-warm-up size	✓ Direct comparison with Sequem in the same tables and metrics	✓ Non-normality: Two queueing processes use distinct service-time distributions. ✓ Autocorrelation/dependence: Two queueing processes contrast dependence patterns. — Nonstationarity. — Model misspecification.	✓ Post-warm-up size evaluated systematically by process, quantile, and regime; time not reported	~ Java package, GUI, and two modes are reported; standalone versions/repository were still under development and replication artifacts are missing

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12	Alexopoulos et al. (2018)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequem for extreme quantiles, revised and compared with Sequest	Extreme quantile; revised precision rule	✓ 1,000 replications in two queues, several quantiles, and two precision regimes, with coverage and post-warm-up size	✓ Direct comparison with Sequest in the same tables and metrics	✓ Non-normality: Queueing processes and quantiles exercise nonnormal forms. ✓ Autocorrelation/dependence: Queueing processes exercise distinct dependence patterns. — Nonstationarity. — Model misspecification.	✓ Post-warm-up size evaluated systematically; total including warm-up and time not reported	~ Mechanism is described, but the complete pseudocode is deferred to 2017 and public software was still under development
13	Alexopoulos et al. (2019a)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequest demonstrated in software; branches: no precision, absolute precision, and relative precision	Stationary quantile; relative/absolute precision	~ One trajectory and several quantiles/configurations; does not estimate coverage	✓ Sequest and Sequem appear in the same scenario/table	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Acquired sizes and precision cost in one realization	✓ Public cross-platform software and interface described
14	Alexopoulos et al. (2019a)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequem for extreme upper quantiles	Extreme upper quantile; sequential precision	~ Results in one trajectory; coverage claims come from earlier works	✓ Compared with Sequest in the same scenario/table	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Sizes observed in the table, without scalability study	✓ Included in selectable public software
15	Alexopoulos et al. (2019b)	Family 1 - Precision or uncertainty sufficiency	Quantile estimation	Sequest architecture; no-precision, absolute-precision, and relative-precision branches	Stationary quantile; relative/absolute precision	~ 1,000 replications and seven problems cover only the no-precision and relative-precision branches; the absolute branch was not evaluated	~ Common comparison between no precision and relative precision; the absolute branch was not compared	✓ Non-normality: Nonnormal forms were deliberately varied. ✓ Autocorrelation/dependence: Dependence structures were deliberately varied. ~ Nonstationarity: Transient behavior provides indirect, limited evidence. — Model misspecification.	✓ Sample size evaluated systematically across problems/quantiles, reaching 57.5 million	✓ Algorithm, GUI, and automated/file modes reported
16	Al-Sumaiti et al. (2019)	Family 4 - Operational-state sufficiency		Moment-based MCS gate in the PV framework; WBD (4 moments), GEVD (2), and enhanced-beta branches	PV output; moment-based gate	— None reported.	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Sample size may vary, but there is no systematic cost/scalability evaluation	~ Tools and principle reported, but stopping and singular cases are not reproducible
17	Bandyopadhyay et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Modified two-stage procedure for a fixed-width CI of log θ	Log odds ratio; two-stage absolute precision	✓ 10,000 repetitions across multiple d/pilots, with substantial undercoverage	✓ Compared with the purely sequential procedure under common scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected sizes/successes evaluated across multiple d and pilots; joint cost not reported	~ Rule described, but variance, critical value, (C,k), and boundaries prevent unambiguous reproduction
18	Bandyopadhyay et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Purely sequential first-passage procedure after each success	Log odds ratio; sequential absolute precision	✓ 10,000 repetitions across multiple d/pilots, generally with higher but still low coverage	✓ Compared with the two-stage procedure under common scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected sizes/successes evaluated across multiple d and pilots; joint cost not reported	~ First-passage rule is explicit, but arm-A acquisition and singular states are unresolved
19	Bandyopadhyay et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Fixed-width CI for $\theta(P)$, two-stage procedure	Cumulative-logit contrast; two-stage absolute precision	✓ Simulation across multiple r, d, and configurations reports coverage and expected size	✓ Compared with the purely sequential mechanism for the same target and scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected sample size evaluated systematically across multiple d/configurations	~ Executable formulas, but no code and incomplete choice of γ
20	Bandyopadhyay et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Fixed-width CI for ρ , two-stage procedure	Stochastic-superiority contrast; two-stage absolute precision	✓ Simulation results across multiple r, d, and configurations are reported for ρ	✓ Compared with the sequential mechanism for ρ in the same scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected size evaluated systematically across multiple scenarios and half-widths	~ Base formulas exist, but variance substitution must be completed and there is no code
21	Bandyopadhyay et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Fixed-width CI for $\theta(P)$, purely sequential procedure	Cumulative-logit contrast; sequential absolute precision	✓ Simulation across multiple r, d, and configurations reports coverage and expected size	✓ Compared with the two-stage procedure for the same target and scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected sample size evaluated systematically across multiple d/configurations	~ Explicit boundary, but no code and no detailed operational cadence
22	Bandyopadhyay et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Fixed-width CI for ρ , purely sequential procedure	Stochastic-superiority contrast; sequential absolute precision	✓ Simulation reports coverage and expected size in multiple scenarios	✓ Compared with the two-stage version of ρ in the same scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected size evaluated across multiple half-widths and configurations	~ Infrastructure reconstructible by analogy, but complete version and code are not supplied
23	Batur et al. (2021)	Family 2 - Discrete-decision sufficiency		QSE - quantile selection with equivalence and set output	Quantile selection; equivalence/set output	✓ PCS and ANO evaluated across multiple distributions, quantiles, δ , numbers of alternatives, and correlations	✓ Comparisons with QS/GQ/LD are reported, with symmetry restrictions explicit	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ ANO evaluated systematically across multiple K, q, δ , and correlations, although its exact unit is NR	✓ Code repository for QSE/QS reported with URL
24	Batur et al. (2021)	Family 2 - Discrete-decision sufficiency		QS - quantile selection without equivalence and unit output	Quantile selection; unit output	✓ PCS and ANO evaluated across multiple distributions, quantiles, δ , alternatives, and correlations	✓ Comparisons with QSE/GQ/LD in common configurations are reported	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ ANO evaluated systematically across multiple sizes/configurations, despite the unit being NR	✓ Code repository for QSE/QS reported with URL
25	Chattopadhyay (2020)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage stratified procedure with SD scale	Stratified mean; SD scale	✓ 10,000 repetitions in two finite populations and two α levels; SD coverage is below nominal in all four cases	✓ Direct comparison with GMD under the same cases, levels, and metrics	✓ Non-normality: Gamma and lognormal distributions were compared. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Sample size/cost quantified in the two populations and two confidence levels	~ Formulas supplied, but code, integerization, fallbacks, and an unambiguous pilot definition are missing
26	Chattopadhyay (2020)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage stratified procedure with GMD scale	Stratified mean; GMD scale	✓ 10,000 repetitions in two populations and two α values; includes severe coverage deterioration in Case 2	✓ Direct comparison with SD under the same cases, levels, and metrics	✓ Non-normality: Gamma and lognormal distributions were compared. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Final size quantified in multiple populations/levels; paired GMD cost not analyzed separately	~ Procedure described by formulas, but without code, calibration, rounding, or fallbacks
27	Cho (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Fixed-width CI for the risk ratio, main two-stage procedure	Risk ratio; two-stage fixed width	✓ 5,000 replications across multiple γ , ϕ , d, and pilots report coverage and E(N)	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ E(N) evaluated systematically across half-widths and pilot sizes; theoretical cost bounds contested	~ Flow and equations given, but no code; rule is undefined in pilot states and has index/rounding ambiguities

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28	Cho et al. (2020)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential fixed-width CI for the risk ratio	Risk ratio; sequential fixed width	✓ 5,000-replicate experiments span several widths, levels and risk ratios, with coverage and sample-size results, including undercoverage	~ Likelihood intervals and reciprocal orientations are compared only narrowly; no fully specified alternative stopping rule is evaluated under a common stopping design	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Sample cost $E(N)$ and oracle n^* are reported across multiple widths/settings	~ Equations identify the procedure, but no code, pseudocode, cap, principled m , or complete boundary handling is supplied
29	Cui et al. (2016)	Family 2 - Discrete-decision sufficiency		Coupled sequential TBD detection and terminal position/trajectory estimation (DO/DPEC/DTEC/DPTEC; regular/truncated branches)	Target detection/trajectory; coupled sequential rule	✓ Monte Carlo evaluates finite SNR ranges, weights and scan counts, with detection, joint classification-failure and ASN outcomes	✓ Sequential variants are compared with four-frame FSS under common simulated radar settings	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. ✓ Model misspecification: Mismatch between design and actual SNR was deliberately varied.	✓ ASN is quantified across hypotheses/SNR settings and compared with the four-frame workload	~ Formulae, flowcharts and recursions identify operations, but code, full executable calibration, time/memory analysis and boundary conventions are unreported
30	Dang et al. (2021)	Family 5 - Resource-aware acquisition and stopping		ALAPI active-learning estimation of a failure-probability function	Failure-probability function; active learning	✓ Three finite computational examples report budgets, call counts and component estimates against IASS/NISS/ASS where available	✓ Identified alternatives are evaluated in shared examples, although Example 3 comparison is restricted to the constant component	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expensive-function call counts are quantified across multiple examples and against alternatives	~ The algorithm and GP setup are described and MATLAB code/model are available only on request; seeds, exact routine and several termination safeguards are unreported
31	De et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Two-stage fixed-width CI for common equicorrelation	Common equicorrelation; two-stage fixed width	✓ Exact calculations and 10^5 -replicate simulations cover multiple k, m, p, α, d settings	✓ The fixed-width rule is compared numerically with transformed bounded-width procedures after scenario-specific calibration	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Expected terminal size and its efficiency relative to n^* are derived/evaluated across multiple parameter and pilot settings	~ Closed-form batch instructions and named numerical integration methods are given, but no code/package or executable supplement is located
32	De et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Transformed bounded-width CIs for common equicorrelation (g1 exact/approximate; g2,g3,g4 adaptive branches)	Common equicorrelation; transformed bounded width	✓ 10^5 -replicate studies vary m, p, δ and compare coverage, size and width across transformations	✓ g1 exact/approximate and adaptive g2/g3/g4 branches are compared under common numerical scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Sample sizes and first-/second-order efficiency are quantified across branches and settings	~ Transformation, horizon and inversion formulas are reproducible conceptually, but no code/package is supplied and the Broyden implementation lacks a function/package locator
33	Dong et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Absolute-precision STS, with STSb/STSs/STSm branches and continuous/grid execution	Steady-state mean; absolute precision	✓ Multiple ϵ , variants, and calibrations in 1,000 M/M/1 replications, including unfavorable results (D-C01-FS)	✓ Sequential quantile versus Student and STSb/STSs/STSm in the same model (D-C01-AC)	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Horizon, time, and storage compared, but without systematic scaling analysis (D-C01-CX)	~ Algorithm 1 only for absolute STSb; code and executable reference not located (D-C01-IM)
34	Dong et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Relative-precision STS extension with multiplicative output	Steady-state mean; magnitude-relative precision	~ One relative experiment with positive target $\alpha=4$; coverage varies with ϵ (D-C02-FS)	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	— None reported.
35	Fitts (2023)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Absolute-width SSR for mean differences, two-group/difference-score branches and $n_{\min}/\alpha$ calibration	Mean difference; absolute precision	✓ Coverage and stopping size evaluated on broad grids and multiple minimum sizes (F-C03-FS)	✓ Common comparison with FSR with/without assurance and nominal monitoring (F-C03-AC)	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Stopping/sample size evaluated on grids, without systematic computational cost (F-C03-CX)	✓ Executable, C code, manual, examples, and R code are reported with locator (F-C03-IM)
36	Flegal et al. (2015)	Family 1 - Precision or uncertainty sufficiency	Cross-cutting	Relative-magnitude fixed-width rule $T_{\cdot}$ with expectation/quantile branches and marginal conjunction	MCMC estimation; magnitude-relative precision	✓ Repeated sequential studies at multiple ϵ , samplers, and targets, shared pairwise across rules (L-C04-FS)	✓ Paired comparison with $T_{\cdot}$ and $T_{\cdot}$ on the same draws (L-C04-AC)	— Non-normality. ~ Autocorrelation/dependence: Distinct chains were used, but autocorrelation was not varied in controlled levels. — Nonstationarity. — Model misspecification.	~ Iterations vary by sampler and tolerance, without systematic resource/scaling evaluation (L-C04-CX)	~ Procedure and packages are named, but complete code, versions, seeds, and KDE details are missing (L-C04-IM)
37	Flegal et al. (2015)	Family 1 - Precision or uncertainty sufficiency	Cross-cutting	Relative-standard-deviation fixed-width rule $T_{\cdot}$ with expectation/quantile branches and marginal conjunction	MCMC estimation; scale-relative precision	✓ Repeated sequential studies at multiple ϵ , samplers, expectations, and quantiles (L-C05-FS)	✓ Paired comparison with $T_{\cdot}$ and $T_{\cdot}$ plus untabulated i.i.d. in one study (L-C05-AC)	— Non-normality. ~ Autocorrelation/dependence: Distinct dependence mechanisms were used, but autocorrelation strength was not controlled. — Nonstationarity. — Model misspecification.	~ Hundreds of thousands of iterations and variation by sampler are reported, without formal scaling (L-C05-CX)	~ Batch means, KDE, and packages are described; bandwidth, options, versions, seeds, and fallback are missing (L-C05-IM)
38	Gong et al. (2016)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	High-dimensional relative FWSR with LCBM, formal/current branches, CBM/LCBM/LCBM(**), and alternative ESS	Joint MCMC estimation; scale-relative precision	✓ 1, 000 stoppings at $p=186$ across multiple thresholds/estimators; fMRI adds a single demonstration (G-C06-FS)	✓ CBM/LCBM/LCBM(**), ESS, and GD compared in a common scenario, with target differences explicit (G-C06-AC)	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Memory evaluated at distinct dimensions/loads and storage order derived; 1.22 MB and 84 MB results reported (G-C06-CX)	~ Complete CBM/LCBM pseudocodes, but scripts do not integrate the source and executed procedure/zero cases are uncertain (G-C06-IM)
39	Gürel et al. (2025)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Standard-error stopping in ACJ, coupled to inherited adaptive acquisition; S-TR40/35/30 and A-TR40-C40 branch	Object abilities; standard-error threshold	✓ 3 sizes $\times$ 3 thresholds, 82 repetitions/cell, plus application	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Object judgments evaluated at three sizes and three thresholds; applied human cost also quantified	~ R environment/packages and ACJ software are mentioned, but code and executable specification are missing
40	Hu (2021)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Multistage normal FWCI family $P(p, k, \epsilon)$, with Stein-type, ACR, Hall, and micro-batch branches	Normal mean; multistage fixed width	✓ 10,000 replications at four d , plus directed condition analysis and illustration	✓ Identified variants compared under common scenarios and metrics	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. ✓ Model misspecification: Two structural conditions were deliberately perturbed.	✓ Operations and sizes evaluated systematically at multiple d, k , and ρ	~ Eqs. (4)-(7) and the described flow provide an equation-reproducible procedure, but no code/package exists and operational details remain incomplete
41	Hu (2022)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	$P(k, p)$: normal fixed-width interval with blocks until fractional gate and unit ACR until stopping	Normal mean; fractional/unit fixed width	✓ 10,000 replications at three d , multiple k, p , plus illustration	✓ Common comparison with ACR ($k=1$) and evidence on operations/size; comparator in groups until the end also analyzed	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Number of operations and size evaluated at multiple d, k, p ; asymptotic difference for groups until the end	✓ Operational rule of $P_{\cdot}$ is reproducible through Eqs. (3.3)-(3.4) and specified output; code/software not supplied
42	Hu (2022)	Family 3 - Risk or loss sufficiency		$P(k, p)$: normal minimum-risk estimation with double-sequential policy	Normal mean; minimum risk	✓ 10,000 replications at three costs c , multiple k, p , plus illustration	✓ Common comparison with ACR ($k=1$) and analysis of comparator in groups until the end	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Operations and size/risk evaluated at multiple c, k, p ; asymptotic size difference supplied	~ Substantial equation-based instructions exist, but Eq. (4.6) prints $T_{\cdot}$ and requires inferred correction $T_{\cdot}$; no code/software

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43	Jiménez Rugama et al. (2018)	Family 4 - Operational-state sufficiency		Variant A for first-order + total effect, with A.a and adaptive A.b/"Correlation 2" branch	Sensitivity indices; first-order and total-effect error bounds	✓ 100 scramblings in two benchmarks and one applied formula, with errors/failures and costs	✓ A.a, A.b, B, and iid/bootstrap comparator evaluated, although iid comparison is confounded	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Cost formula and calls to f evaluated in several problems; A/A.b depend on dimension and horizons	~ Algorithm 1 and equations provide partial pseudocode/procedure, but no code/package; singular states and A.b catch-up lack a rule
44	Jiménez Rugama et al. (2018)	Family 4 - Operational-state sufficiency		Variant B using replicated designs, first-order indices only	Sensitivity indices; first-order error bounds	✓ 100 scramblings in two benchmarks and one applied formula, with errors/failures and costs	✓ Compared with A.a/A.b under the same problems and tolerance	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Cost $2 \cdot 2^{\{m\}}$ formally independent of d and empirically reported in several problems	~ Algorithm 1/equations describe the procedure, but no code/package and rules for invalid finite states are missing
45	Karmakar et al. (2019)	Family 3 - Risk or loss sufficiency		Sequential plug-in risk-efficient rule to estimate the fixed part of multivariate RCA	Multivariate RCA; plug-in risk	✓ Monte Carlo at multiple horizons and configurations	✓ Common comparison with fixed oracle	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	— None reported.
46	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential AIPE for Pearson with asymptotic Lee interval	Pearson correlation; absolute precision	✓ Multiple levels, widths, correlations, and DGPs; includes undercoverage	✓ Compared with two alternative Pearson intervals	~ Non-normality: Only one gamma condition was evaluated beyond normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R/MBESS function demonstrated, without complete inventory
47	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Pearson-Corty-Corty AIPE	Pearson correlation; absolute precision	✓ Simulated in multiple normal and gamma conditions	✓ Compared with Pearson-Lee and Moinester-Gottfried	~ Non-normality: Only one gamma condition was evaluated beyond normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code for methods is declared, without mapping by variant
48	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Pearson-Moinester-Gottfried AIPE	Pearson correlation; absolute precision	✓ Multiple finite conditions, but only normal	✓ Compared with Pearson-Lee and Corty-Corty	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code declared, without mapping by variant
49	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential AIPE for Kendall τ	Kendall tau; absolute precision	✓ Multiple normal and gamma conditions	— None reported.	~ Non-normality: Only one gamma condition was evaluated beyond normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code declared, without mapping by variant
50	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential AIPE for Spearman ρ_s	Spearman correlation; absolute precision	✓ Multiple normal and gamma conditions	— None reported.	~ Non-normality: Only one gamma condition was evaluated beyond normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code declared, without mapping by variant
51	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential AIPE for ρ^2 - Bonett-Wright	Squared multiple correlation; absolute precision	✓ Multiple k, ρ^2 , widths, and levels	✓ Common comparison with Olkin-Finn	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code declared, without mapping by variant
52	Kelley et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Sequential AIPE for ρ^2 - Olkin-Finn	Squared multiple correlation; absolute precision	✓ Multiple k, ρ^2 , widths, and levels	✓ Common comparison with Bonett-Wright	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ R code declared, without mapping by variant
53	Hwang (2023)	Family 3 - Risk or loss sufficiency		Robust two-stage procedure ($U_{c,X}(U_c)$)	Vector mean; two-stage sampling-cost trade-off	✓ Seven costs, three true priors, and two loss matrices	✓ Compared with correct/misspecified N_c and A.P.O.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	— None reported.
54	Maurya et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Two-stage NP - upper one-sided SCI	Treatment-control contrasts; simultaneous upper intervals	✓ Grid of n, δ , k, and scales, 30,000 replications	✓ Paired comparison with LN	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ Supplementary R code only to calculate d, not a full implementation evaluated
55	Maurya et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Two-stage NP - lower one-sided SCI	Treatment-control contrasts; simultaneous lower intervals	✓ Grid of n, δ , k, and scales, 30,000 replications	✓ Paired comparison with LN	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ Supplementary R code only to calculate d, not a full implementation evaluated
56	Maurya et al. (2017)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Two-stage NP - two-sided SCI and directional decision	Treatment-control contrasts; simultaneous bilateral intervals	✓ Coverage/volume on a broad grid, without direct estimate of $b(\delta)$	✓ Paired comparison with LN	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ Supplementary R code only for d; full implementation not evaluated
57	Mukherjee et al. (2021)	Family 3 - Risk or loss sufficiency		Cost-aware active estimation (CAE): posterior mean, Fisher-information plug-in acquisition, and posterior-risk stopping; K=1 branches and K=2 illustration	Posterior estimate; posterior-risk stopping	~ One synthetic illustration across several β , without complete Monte Carlo protocol	✓ CAE versus random selection and method [19] in the same graph and metric	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ General cost qualitative (posterior integration and information of all sources), without scaling analysis	— None reported.
58	Mukhopadhyay et al. (2016)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Purely sequential fixed-width CI for θ ; truncated J branch and open extensions	Scalar parameter; sequential fixed width	✓ Monte Carlo in multiple configurations, with observed finite undercoverage	~ Compares (m, ϵ) and I/J configurations; J is not a probabilistically distinct alternative	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Online update by two sums; sample cost analyzed, without scalable computational benchmark	— None reported.
59	Mukhopadhyay et al. (2016)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Multiplicative fixed-accuracy CI for θ ; exact-horizon and approximations 1-4 branches	Scalar parameter; multiplicative fixed accuracy	✓ Calculations and Monte Carlo in multiple configurations	✓ Exact horizon and approximations compared on common grids	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Calculated sizes and quantiles discussed, without computational scalability study	— None reported.
60	Mukhopadhyay et al. (2016)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Bounded-accuracy CI for $\tau(\theta)$, with exact/approximate-horizon branches	Transformed parameter; bounded accuracy	✓ Monte Carlo at multiple θ values/configurations	~ Exact/approximate horizons are considered, but direct evaluation of L_n is narrow	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Fixed horizon and closed-form formula; no resource-scaling evaluation	— None reported.

ID	Study	Family	Subfamily	Operational contribution and branches	Consider when	Finite-sample evidence	Alternative comparison	Robustness	Computational / scalability evidence	Implementation availability
61	Mukhopadhyay et al. (2018)	Family 3 - Risk or loss sufficiency		Two-stage normal MRPE; corrected-sample, GMD, and MAD branches	Normal mean; two-stage minimum risk	✓ 10, 000 replications at four n^* and three σ_L	✓ Three new branches compared in the same design; no direct comparison with antecedents	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Naive GMD may cost more, but no scaling analysis or measurement	— None reported.
62	Mukhopadhyay et al. (2018)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage normal FCIE; corrected sample-variance, GMD ² , and MAD ² branches	Normal mean; two-stage absolute precision	✓ 10,000 replications at five C, with undercoverage at the smallest sizes	✓ Three new branches compared in the same design; previous Student/pivotal rules are not empirical comparators	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Naive GMD may cost more, but no scaling analysis or measurement	— None reported.
63	Mukhopadhyay (2020)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	P1: exponential fixed-width CI centered on all accumulated data	Exponential mean; fixed width	✓ 5,000 replications at six C values, with reported coverages	✓ Common comparison with P2 and P3 at the same stopping times	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Online rule with sum/mean and sample cost analyzed; no computational scalability test	— None reported.
64	Mukhopadhyay (2020)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	P2/P3: exponential CIs centered on the stopping time; P3 adds bias correction	Exponential mean; stopping-time CI	✓ 5, 000 replications at six C; separable results for P2/P3	✓ P2, P3, and P1 compared at the same N_d	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Output is computationally closed after stopping; no systematic resource evaluation	— None reported.
65	Mukhopadhyay et al. (2020a)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	V1 - nonparametric sequential FWCi for the mean	Nonparametric mean; fixed width	✓ multiple pilots, benchmarks, and distributions	— None reported.	✓ Non-normality: Multiple nonnormal distributions and severity levels were evaluated. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Final size evaluated across multiple benchmarks	~ Executable rule described; code/seed not reported
66	Mukhopadhyay et al. (2020a)	Family 3 - Risk or loss sufficiency		V2 - nonparametric sequential MRPE for the mean	Nonparametric mean; minimum risk	✓ multiple pilots, benchmarks, and distributions	— None reported.	✓ Non-normality: Nonnormal distributions and mixtures were compared. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Final size evaluated across multiple benchmarks	~ Executable rule described; code/seed not reported
67	Mukhopadhyay et al. (2020a)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	V3 - nonparametric sequential FWCi for variance	Nonparametric variance; fixed width	✓ multiple pilots, benchmarks, and populations	— None reported.	✓ Non-normality: The distributional stress set includes t(8) scenarios. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Final size evaluated across multiple benchmarks	~ Procedure described; no code and no correction for negative lower limit
68	Mukhopadhyay et al. (2020a)	Family 3 - Risk or loss sufficiency		V4 - nonparametric sequential MRPE for variance	Nonparametric variance; minimum risk	✓ multiple pilots, benchmarks, and populations	— None reported.	✓ Non-normality: The distributional stress set includes t(8) scenarios. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Final size evaluated across multiple benchmarks	~ Procedure described; code/seed not reported
69	Mukhopadhyay et al. (2020b)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Helmert class $N_{\{r,h\}}$ - FWCi branch	Normal mean; Helmert fixed width	✓ 10,000 replications, multiple r, and benchmarks	✓ r branches compared; $r=2$ optimal only in internal asymptotic variance	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Size/excess evaluated from 50 to 2, 000	~ Clear online update; constants/code/seed incomplete
70	Mukhopadhyay et al. (2020b)	Family 3 - Risk or loss sufficiency		Helmert class $N_{\{r,h\}}$ - MRPE branch	Normal mean; Helmert minimum risk	✓ 10,000 replications, multiple r, and benchmarks	✓ r branches compared; $r=2$ minimizes the pertinent asymptotic term	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ size, risk, and regret evaluated from 50 to 2,000	~ Clear online update; constants/code/seed incomplete
71	Mukhopadhyay et al. (2020c)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	FWCi for mean with unit and batched acquisition; $P_{\{1,k\}}$, $P_{\{2,k\}}$ branches	Mean; unit/batched fixed width	✓ 10,000 runs at multiple k, pilots, and oracles; includes undercoverage	✓ Branches and usual procedure compared in the same model	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Terminal size evaluated at multiple k and oracles	~ Implementable rules and MATLAB reported; code not supplied
72	Mukhopadhyay et al. (2020c)	Family 3 - Risk or loss sufficiency		MRPE for mean with unit and batched acquisition; $Q_{\{1,k\}}$, $Q_{\{2,k\}}$ branches	Mean; unit/batched minimum risk	✓ 10,000 runs at multiple k, pilots, and oracles	✓ Branches and usual procedure compared in the same model	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Size/risk evaluated at multiple k and oracles	~ Implementable rules and MATLAB reported; code not supplied
73	Mukhopadhyay et al. (2020d)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Batched FWCi with permuted estimators $P\#_{\{1,k\}}$, $P\#_{\{2,k\}}$	Mean; batched permuted fixed width	✓ 10,000 runs at multiple k, pilots, and targets	✓ New branches versus previous batched ones under common scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Size and dispersion evaluated at multiple k and C_k	~ Accumulators and rule described; code/seed not reported
74	Mukhopadhyay et al. (2020d)	Family 3 - Risk or loss sufficiency		Batched MRPE with permuted estimators $Q\#_{\{1,k\}}$, $Q\#_{\{2,k\}}$	Mean; batched permuted minimum risk	✓ 10,000 runs at multiple k, pilots, and targets	✓ New branches versus previous batched ones under common scenarios	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ size, risk, and dispersion evaluated at multiple k and n_k^*	~ Accumulators and rule described; code/seed not reported
75	Mukhopadhyay et al. (2022)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	MRFWCi formulation and unified theory for normal-mean FWCi, with purely sequential, accelerated, three-stage, and lower-bound two-stage branches	Normal mean; multi-stage fixed width	✓ Simulation at seven oracle sizes, 10,000 replications, and coverage/risk metrics; application is retrospective	✓ Four rules compared under the same population, targets, and metrics	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Comparative times in one workload, without scaling evaluation or reproducible environment	— None reported.
76	Mukhopadhyay et al. (2024)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Unified second-order theory for FWCi of contrast between two NE locations, with P1-P4 branches	Two-location contrast; fixed width	✓ Monte Carlo over an ideal-size range and stated parameter grid, with 100,000 repetitions; published results are parametrically narrow	✓ P1-P4 compared in the same problem, scenarios, and metrics	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Computational time evaluated along ideal sizes/workloads and across strategies, with environment reported	— None reported.
77	Park (2016)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Numerical comparison of sequential estimation under covariate-adjusted CARA versus RA described as unadjusted, with precision stopping	Joint parameter estimation; confidence-region precision	✓ Simulations across multiple pilots and tunings report stopping, coverage, and CAP, including unfavorable results and exceptions	~ RA comparator exists, but it is unclear whether only allocation or also model, estimand, contrast, and stopping covariance change	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Sample resource/stopping time quantified systematically across multiple pilots and configurations	— None reported.

ID	Study	Family	Subfamily	Operational contribution and branches	Consider when	Finite-sample evidence	Alternative comparison	Robustness	Computational / scalability evidence	Implementation availability
78	Quan et al. (2025)	Family 5 - Resource-aware acquisition and stopping		Sequential Bayesian estimation with precision-ordered transmission and posterior-width stopping	Posterior estimate; posterior-width precision	~ Synthetic evaluation uses N=50 in one configuration; metrics and results exist, but size, α , and Monte Carlo are incomplete	✓ Informed order, random order, and all sensors compared in common configurations by MSE and transmissions	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Mean transmission count is quantified/limited and compared, but no systematic scaling or complete energy model	— None reported.
79	Sajjadipanah et al. (2021)	Family 3 - Risk or loss sufficiency		Two-stage point estimation of β in AR(1); purely sequential comparator branch	AR(1) coefficient; two-stage point risk	✓ Monte Carlo at four (m,A) pairs, a β grid, and 10,000 replications	✓ Common comparison with the purely sequential rule by size, risk, and RMSE	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Lower update frequency is qualitative; cost not measured	~ Executable formula, but no code/pseudocode and finite states without fallback
80	Sajjadipanah et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Other scalar-parameter estimation	Two-stage fixed-width CI for β ; purely sequential comparator branch	AR(1) coefficient; two-stage fixed width	✓ Coverage and size evaluated across multiple pilots, widths, and β values	✓ Comparison in the same design with the purely sequential interval rule	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ One-additional-batch architecture is qualitatively simple; no scaling evaluation	~ Explicit rule, but ambiguous normal critical value, nonpositive plug-in, and singular denominator without safeguard
81	Schmalofski et al. (2023)	Family 4 - Operational-state sufficiency		Joint terminal criterion: interpolated energy stability and persistence of N_cusps, with 1D/2D branches	Energy/cusp structure; joint stability	✓ Trajectories at multiple thresholds/configurations in 1D and four 2D cases, with metrics and results	~ Comparison with regular grid evaluates acquisition+stopping jointly, without stopping ablation	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Discusses grid cost s^2 and simulation reduction, without systematic scalability study	~ Partial flow and rules; code in development only on request and supplement absent
82	Schmalofski et al. (2023)	Family 4 - Operational-state sufficiency		2D extension of sequential design and minima/valley detector	Two-dimensional minimum/valley; structural stability	✓ Four 2D landscapes with stated budgets and results against references	✓ Combined procedure compared with regular grids of the same total budget	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Grid construction s^2 is said to be costly, without quantified scalability	~ Grid, 8-connectivity, and consolidation are described; edges/ties and query modality remain uncertain
83	Solanky et al. (2015)	Family 2 - Discrete-decision sufficiency		Single-stage ternary classification, known variance, with two symmetric indifference zones	Ternary classification; known-variance indifference zones	— None reported.	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Size grows as $1/(ra)^2$ and b requires numerical integration; no scaling test	~ Formulas and b table, but no code, pseudocode, or tie-breaking rule
84	Solanky et al. (2015)	Family 2 - Discrete-decision sufficiency		Ternary classification with global purely sequential stopping under unknown variance	Ternary classification; unknown-variance sequential rule	✓ Simulation at multiple targets n^* and r, 5,000 repetitions, PCD and E(N)	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Relation $N \approx n^*$ and $1/(ra)^2$ growth formalized, but without scaled computational cost	~ Explicit mathematical rule; no code, tolerances, cap, or tie-breaking
85	Steland (2017)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage normal fixed-width VWA CI	Weighted mean; two-stage absolute precision	✓ Coverage and size evaluated at multiple d, n, and locations; part of reporting is incomplete	✓ Common comparison with bootstrap calibration in Tables 6-9, with mixed results	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Horizon and O(n) cost described; no systematic scalability evaluation	~ Explicit rule, but second-batch count is inconsistent and integer operators/cap are undefined
86	Steland (2017)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage fixed-width VWA CI with studentized/smoothed bootstrap quantile	Weighted mean; bootstrap-calibrated precision	✓ Tables 8-9 use 50,000 runs, B=2000, and multiple d/locations; improvement is nonuniform	✓ Comparison with normal variant under common scenarios and coverage	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Bootstrap is described as more demanding, without times or scalability	~ Partial formulas; smoothing width, rounding, degenerate studentization, and cap unresolved; no code
87	Steland et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Generic nonparametric two-stage fixed-width interval rule; high-accuracy/high-confidence regimes and first-stage branches	Mean estimation; two-stage absolute precision	✓ Simulations cover multiple d, levels, distributions, and initial minima, including unfavorable results	✓ Minima 15/30 are compared in the same design; alternative branch is also formalized	✓ Non-normality: Normal, t(5), and uniform distributions were compared. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Sample efficiency is formalized and sizes measured, but no computational scalability evaluation	~ Algorithm and formulas are reproducible at a high level; code/package not reported
88	Steland et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Extension to increasing-dimension data projections	Projected mean; increasing dimension	— None reported.	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Conditions allow growing p_n, but cost/scalability are not quantified	~ Formulas for procedure and variance are given; no executable artifact
89	Steland et al. (2021)	Family 1 - Precision or uncertainty sufficiency	Mean estimation	Two-stage rule for common mean of two samples with random weights	Common mean; two-sample random weights	✓ Table 2 evaluates multiple d, levels, and distributions, although it contains internal inconsistencies and difficult performance under t(5)	~ There are horizon/variant comparators in the same study, but unit and truncation prevent fully coherent reading	✓ Non-normality: Normal, t(5), and uniform distributions were compared. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Sizes/oversizing are reported, but cost unit is uncertain and there is no computational scaling	~ Formulas and plug-in are supplied, but no code and no fallback for 0/0
90	Tavana et al. (2015)	Family 4 - Operational-state sufficiency		Integrated entropic framework of synthetic extrapolation, EV ranking, and validation/continuation gate	EV ranking; validation/continuation gate	✓ One/two loops, four configurations, and Monte Carlo with two and ten evaluations are evaluated with rankings/results	✓ Uniform/normal/PERT Monte Carlo and four variants of two bounded algorithms compared in the common case	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. ~ Model misspecification: Alternative domains were evaluated, but as a single limited perturbation.	~ Growing intractability is recognized and motivates expected plug-in; no times, orders, or systematic scaling	~ Equations and flowchart are supplied, but code, complete numerical method, and fallbacks are missing
91	Wu et al. (2021)	Family 4 - Operational-state sufficiency		Precision analysis under uncertain current: model, relative-error stopping MCS, surfaces, and Sobol	Motion-error mean; relative-error threshold	✓ Twenty configurations, MCS with 1,950 realizations, and multiple results/sensitivities provide finite evaluation	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Counts (20, 30,000, 1,950) and MATLAB integration are reported, without systematic scaling study	~ Equations, integrator, parameters, and polynomials are reported; code, seeds, and complete protocol are not
92	Wu et al. (2021)	Family 4 - Operational-state sufficiency		Intermittent yaw-feedback strategy by calculated duration	Yaw response; duration condition	~ Evaluated by simulation with and without feedback, but in one illustrated configuration and without prospective physical trial	✓ Direct comparison with no-feedback regime uses $\mu_D/\Delta\mu_D$ in the same model	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ Logic, parameters, and actuation are described, but ramps, saturation, division by zero, and "End control?" remain incomplete
93	Zhao et al. (2021)	Family 5 - Resource-aware acquisition and stopping		CEMHMB for reliability evaluation: CE-SS-M-H, minimum-burden stopping, and terminal IS	Reliability estimation; computational burden and probability precision	✓ Multiple systems/cases and p/y sensitivities report burden, time, LOLP, and EENS	✓ CEIS, CMCS, and pure SS compared in common scenarios with cost and reliability metrics	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	✓ Burden and time quantified in several systems/workloads, with environment and budgets reported	~ Three algorithms and equations supplied, but no code and finite states/terminal formula unresolved
94	Zhuang et al. (2021)	Family 3 - Risk or loss sufficiency		Minimum-risk point estimation of population size, with pure, conventional accelerated, and practical batched accelerated branches	Population size; minimum risk	✓ Simulations in multiple configurations for pure and batch branches, with metrics and results; scopes separate	~ Comparison with limited-risk method, but without common objective function	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	~ Costs/samples are theoretically quantified and compared, but there is no systematic evaluation of computational cost or operational scalability	~ Formulas and steps are partly reproducible; no code and rules for infeasible states, rounding, and update convention are missing

ID	Study	Family	Subfamily	Operational contribution and branches	Consider when	Finite-sample evidence	Alternative comparison	Robustness	Computational / scalability evidence	Implementation availability
95	Zoubeidi et al. (2019)	Family 1 - Precision or uncertainty sufficiency	Joint parameter estimation	Fixed-size sequential regions for θ and $\gamma=A\theta$, with precision/N grid extension	Joint parameter estimation; confidence-region precision	✓ Simulations in 1,000 repetitions, multiple designs/coefficient values, with reported coverage, including unfavorable results	— None reported.	— Non-normality. — Autocorrelation/dependence. — Nonstationarity. — Model misspecification.	— None reported.	~ Software and numerical operations are cited, but no executable artifact or failure handling exists and there is a sign inconsistency in information